

SYED MUHAMMAD RAFAY ALI

Full Stack ML Engineer

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PROFESSIONAL SUMMARY

Full-stack AI engineer and Computer Science graduate from FAST-NUCES with hands-on experience building full-stack web applications and AI-powered systems using React, Next.js, TypeScript, Node.js, Django and Python. Proven ability to design and deploy RESTful APIs, train and serve computer vision models with YOLO, PyTorch, and OpenCV, and build CI/CD pipelines with GitHub Actions and Docker across five end-to-end projects spanning booking platforms, AI-powered event management, and real-time computer vision systems. Additional exposure to machine learning techniques, including neural networks and K-Means clustering, applied to real-world route optimization and object detection problems.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES) 2022 – 2026
Bachelor of Science in Computer Science

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python

Frontend: React, Next.js, React Router, TanStack Query, HTML5, CSS3, Tailwind CSS, Bootstrap, Vite, Cross-Platform Compatibility

Backend: Node.js, Express.js, FastAPI, RESTful APIs, JWT, OAuth, Django

Machine Learning / Computer Vision: PyTorch, OpenCV, YOLOv11 (Ultralytics), scikit-learn, Neural Networks, K-Means Clustering, Kaggle

Databases & DevOps: MongoDB, PostgreSQL, Supabase, Docker, GitHub Actions, CI/CD, Git, Vercel, Jest, Supertest

CERTIFICATIONS

[Meta Full Stack Developer: Front-End & Back-End from Scratch | Meta / Coursera](#)
[Prompt Engineering | Coursera](#)

PROJECTS

HostelSpot: Student Hostel Booking Platform | *React, TypeScript, Node.js, Supabase (PostgreSQL), Tailwind CSS, Docker, GitHub Actions*

- Built a full-stack hostel booking platform with dual dashboards for students and owners, real-time availability updates, and an admin panel; designed a responsive, cross-browser UI.
- Developed RESTful APIs integrated with Supabase (PostgreSQL) and crafted data models for authentication and real-time data sync, reducing booking confirmation time by ~60%.
- Configured a CI/CD pipeline using GitHub Actions and Docker to automate testing and deployment.

ELDTrack: FMCSA-Compliant ELD Trip Planner | *React, Vite, Leaflet, Django, Django REST Framework, OSRM, Nominatim*

- Built a full-stack HOS compliance engine that computes FMCSA-regulated driving limits (11-hour/14-hour/70-hour rules) and automatically inserts mandatory breaks, 34-hour restarts, and fuel stops for multi-day route plans.
- Integrated OSRM and Nominatim APIs for route geocoding and distance/duration calculation, rendering interactive route and stop visualizations with React-Leaflet.
- Dynamically generated multi-day FMCSA-standard daily log graphs in SVG and validated the isolated HOS engine with a Django test suite covering short trips, long hauls, and max-cycle restart scenarios.

CarInsight Pro: Car Research & Comparison Platform | *React, Node.js, Express, MongoDB, GitHub Actions, Jest, Supertest*

- Built a MERN-stack platform for side-by-side car specification comparisons with an interactive matchmaking quiz and a custom CMS.
- Designed RESTful API routes serving structured car data with filtering, sorting, and comparison queries; ran 7 Jest/Supertest integration tests against an in-memory MongoDB instance on every push before deployment to Railway.

PakTravel AI: Intercity Bus Travel System | *Python, Pandas, NumPy, scikit-learn, Neural Networks, K-Means Clustering*

- Built an AI-powered intercity bus travel system combining classical search algorithms, propositional logic, and CSP-based scheduling to model real-world route constraints.
- Applied neural networks and K-Means clustering for route optimization and demand analysis, processing structured travel datasets with Pandas and NumPy to train and validate models across multiple route configurations.

HelmetGuard AI: Real-Time Motorcycle Helmet Detection System | *Next.js, FastAPI, YOLOv11, OpenCV, PyTorch, Docker, Kaggle*

- Built a full-stack computer vision system that detects whether motorcycle riders are wearing helmets from uploaded images/video or a live webcam feed, using a fine-tuned YOLOv11 model to classify person, helmet, and no-helmet with bounding boxes and confidence scores.
- Trained the detection model on Kaggle using a dataset of 8,000+ labeled images over 60 epochs, with automatic checkpoint resumption for interrupted training runs.
- Developed a FastAPI backend for real-time inference and a Next.js dashboard displaying live detections, an adjustable confidence threshold, and aggregate helmet-compliance statistics.
- Containerized the frontend and backend with Docker Compose for one-command setup and reproducible local development.